RAVI TEJA CHIMATA
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EXPERIENCE

- Over 8 years of experience specializing in **Quality Engineering and Supplier Quality Management**, with strong proficiency in Advanced Product Quality Planning(APQP), Root Cause Analysis (RCA), Problem Solving, CAPA, and Process/Layered Process Audits (LPA), driving continuous improvement(Kaizen) across manufacturing operations.
- **Core Skills: APQP, PPAP, RCA, PFMEA, Control Plans, DVP&R, DOE, SPC, MSA, GD&T, Audit Management, Risk Mitigation**
- Broad commodity experience across mechanical, electrical, interior, exterior, functional, and structural automotive systems, with deep expertise in EV battery modules, plastic injection molding components,General Assembly,EOL Testing and BIW stamping.
- Strong collaboration with customers, suppliers, design engineering, and cross-functional teams to drive corrective actions, defect reduction, continuous improvement, and overall quality performance.
- Proven success in leading and mentoring Quality Technicians, Inspectors, and Supervisors during new program launches, including gage training, non-conformance management**, sorting activities, and MRB reviews**.
- Demonstrated success in deploying and improving quality management systems aligned with IATF 16949, ISO 9001, and AIAG standards, ensuring compliance, audit readiness, and operational excellence.
- Experience working with fast paced OEMs such as Tesla and Toyota, with hands-on use of supplier portals, PLM systems, and Plex Smart Manufacturing platforms.
- Deep manufacturing process knowledge across Injection Molding, Roll Forming, Extrusion, BIW Stamping (Transfer & Progressive), Welding (MIG/TIG/RSW), Heat Treat, Machining, Assembly, and Coating operations (Powder Coat & E-Coat).

Achievement's: -

- Successfully launched **15+ new model programs** from design to mass production with **zero major quality issues**, using APQP and stringent project management controls.
- Achieved **first-pass PPAP approvals** for multiple product launches with **zero customer escalations**.
- Executed and documented **100+ non-conformance investigations** using 8D and CAPA, driving effective root-cause elimination.
- Reviewed **50+ engineering drawings** using GD&T to establish inspection standards and ensure manufacturability.
- Implemented **50+ Process Change Requests (PCRs)** driven by ECIs/ECRs and supplier changes, validating process robustness through capability and data analysis.
- Conducted **MSA and Gage R&R** studies that improved powder-coat thickness measurement repeatability by deploying precision digital gauges and gage blocks.
- Led a major **continuous improvement project** cutting exterior defect rates from **10% to 4%**, saving **$15,000 annually** and reducing repetitive inspection effort by **30%**.
- Eliminated slug-mark and dimensional variation issues in stamping, delivering **$8,000 scrap savings**, **12% defect reduction**, **5% OEE improvement**, and longer tool life.
- Strengthened **Poka-Yoke systems**, reducing customer issues by **80%**, saving **$20,000** in sorting costs, and lowering overall defects by **70%**.
- Led multiple **root-cause investigations** using 8D, Fishbone, 5-Why, and DMAIC, driving corrective actions and process optimization.
- Improved manufacturing line capability with Cpk / Ppk increases **from <1.0 to >1.33**, ensuring customer compliance and audit success.
- Successfully managed **RMA and customer escalations**, implementing clear 8D reports and improving QBR satisfaction scores.
- Conducted internal, supplier, and **CQI Special Process Audits** (CQI-9, CQI-11, CQI-12, CQI-15, CQI-23), strengthening process control and IATF 16949 compliance.
- Recognized for a hands-on, data-driven leadership style and strong shop-floor presence, enabling fast issue resolution and cross-functional alignment.
- Supported installation and **qualification of new injection molding and stamping tooling**, ensuring process capability from day one.
- Ensured compliance with **IATF 16949, ISO 9001, ISO 14001**, and customer-specific requirements during audits with zero major nonconformances.

YANFENG– HAYWARD, CA, USA
LEAD QUALITY ENGINEER - Automotive Interiors (Floor Console - Tesla Model 3, Model Y) JUL 2024 – DEC2024

- Prepared and delivered complete Launch PPAP packages, ensuring full customer compliance, documentation accuracy, and production readiness and Solved problems under pressure for fast paced environment.
- Supported appearance quality evaluations, AAR (Appearance Approval Process), and managed Class A surface quality concerns for floor console assemblies.
- Conducted initial process capability studies (Cp/Cpk), validated DV/PV builds, and verified dimensional, functional, and performance compliance to customer specifications.
- Reviewed engineering drawings using GD&T principles to define inspection criteria, ensure manufacturability, and support the development of robust measurement methods.
- Managed end-to-end supplier quality activities, including supplier selection, technical reviews, kickoff meetings, process approvals, PPAP submissions, and corrective actions.
- Supported development and maintenance of Design & Process FMEAs, Characteristic Matrices, Control Plans, and LPA deployment to strengthen preventive controls.
- Supported **Class A surface** evaluations **and textile edge-wrapping** quality for interior trim, ensuring flawless appearance, seam alignment, wrap adhesion, and overall customer-visible finish.
- Owned and managed Launch-phase 8D corrective actions across customers, internal manufacturing, and the supplier base, ensuring rapid containment and recurrence prevention.
- Managed supplier sort initiation and closure rates coordinated Clean Shipment Dates (CSD) and ensured SCAR Verification of Effectiveness (VOE) while tracking supplier KPIs such as aging MRB tickets to improve supplier accountability.

TESLA INC– SPARKS, NV, USA
Sr. QUALITY ENGINEER - EV Battery Module Components (Tesla Model 3, Model Y) JAN 2024 – JUN 2024

- Developed the manufacturing requirements, quality standards, and product quality plans to ensure products will meet the expectations of our target customers.
- Reviewed engineering drawings using GD&T principles to establish inspection criteria, validate manufacturability, and support robust production plans.
- Managed quality for EV battery subsystems including coolant tubes, battery cells, stamping press parts, and injection-molded clamshell components used in HV battery modules.
- Translated subsystem and component requirements into clear manufacturing and inspection expectations, aligning production with design intent and critical-to-quality (CTQ) features.
- Identified and prioritized critical requirements to direct supplier and internal focus toward high-risk characteristics and failure modes for RPN evaluation in PFMEA.
- Led cross-functional Design & Process FMEA sessions to identify product and process risks and implement effective mitigation strategies for Kaizen and Gemba activities.
- Applied statistical tolerancing and capability studies (Cp/Cpk, Pp/Ppk) to evaluate variability impacts and validate robust design margins.
- Managed root cause investigations and corrective actions using structured problem-solving methods (RCA, 8D, 5-Why, Fishbone), ensuring sustainable preventive actions.

AISIN USA MFG., INC– SEYMOUR, IN, USA
QUALITY ENGINEER – Automotive Exteriors, Body, Functional, Stamping Products MAR 2018 – DEC 2023
NEW MODEL LAUNCH | SUPPLIER QUALITY | CHANGE MANAGEMENT | PLANT QUALITY MASS PRODUCTION |

Advanced Product Quality Planning (APQP)

- Managed end-to-end lifecycle quality for new automotive exterior and body components from concept through mass production.
- Developed and maintained PFMEAs, Control Plans, Process Flow and APQP core documentation for new platforms.
- Coordinated with design engineering teams to support **APQP gate reviews** by evaluating **DVP&R**, validating design robustness, and ensuring readiness for prototype, pilot, and production builds.

Internal Plant Quality Management

- Managed end-to-end customer quality concerns, including defect verification, sorting, short-term countermeasures, and detailed 8D analysis, ensuring timely implementation of long-term corrective actions and improved customer satisfaction.
- Implemented preventive measures including PFMEA updates, error-proofing (Poka Yoke), and process improvements to reduce defects, scrap, and rework.

- Led inspection teams across incoming, in-process, and final operations, utilizing gauges, fixtures, and CMMs for dimensional verification.

Supplier Quality Management

- Reviewed and approved supplier **PPAP submissions, SCARs**, and corrective actions, ensuring adherence to customer and internal standards.
- Conducted supplier audits, monitored **process capability (Cp/Cpk/Pp/Ppk),** and drove 8D problem-solving and continuous improvement initiatives.
- Managed supplier qualification, incoming inspection standards, and Tier 2/Tier 3 supplier performance metrics.

Auditing & Compliance

- Performed internal and supplier audits to ensure adherence to **AIAG guidelines, IATF 16949** requirements, and internal QMS processes.
- Maintained accurate quality documentation, trend analysis, deviation summaries, and management reporting.

Field & Warranty Analysis

- Conducted **warranty returns analysis**, failure mode investigations, and root cause analysis for field issues.
- Implemented supplier chargebacks and collaborated with cross-functional teams to reduce warranty costs and enhance customer satisfaction.

Root Cause Analysis & Continuous Improvement

- Led structured problem-solving initiatives using **8D, 5 Why, Ishikawa, and DMAIC methodologies**.
- Executed corrective and preventive actions to eliminate recurring defects, improve process stability, and optimize production efficiency.

Change Management & Product Updates

- Evaluated engineering changes for design or process updates, ensuring minimal production impact and proper validation.
- Implemented engineering **change instructions (ECIs, ECRs & PCR's)** and updated associated documentation in alignment with QMS standards. Supported continuous improvement, standardization, and Lean Manufacturing initiatives.

Process Capability & Statistical Control

- Performed process capability studies and statistical process control (SPC) monitoring to ensure production consistency.
- Analyzed production and quality data to drive PPM reduction, scrap minimization, and operational efficiency improvements.

Training & Measurement Expertise

- Conducted operator training programs on inspection techniques, measurement systems, and gauge usage.
- Led Gage R&R studies to validate measurement systems and ensure reliable inspection outcomes.

Customer Quality & Escalation Management

- Strong customer-facing experience handling escalations, **8D submissions**, and on-site containment with OEM quality teams. Managing Quality Problem Reports SCAR, (QPRs/QIRs) .

Product Reliability & Durability Testing

- Supporting lab testing, environmental testing, and durability studies.
- Ensuring parts meet functional performance over lifecycle.

Cost of Poor Quality (COPQ) Management

- Leading initiatives to quantify, reduce, and prevent costs associated with defects, warranty, scrap, and rework.
- Identified nonconformances and implemented corrective actions to improve operational efficiency and reduce **Cost of Poor Quality (COPQ).**

Lean, Six Sigma, & Operational Excellence

- Driving process improvements using Lean, Kaizen, Six Sigma, and DMAIC methodologies.
- Supporting standard work, line optimization, and cycle time reduction.

EDUCATION

MASTER OF SCIENCE in INDUSTRIAL ENGINEERING- Wayne state university MAY 2017

BACHELOR OF MECHANICAL ENGINEERING - (VIT) Vellore, India JUN 2014

ADDITIONAL

CORE COMPETENCIES

APQP • PPAP • PFMEA • Control Plans • DVP&R • DOE • SPC • MSA • GD&T • 8D • DMAIC • LPA • Process Audits • CQI special process audits • Risk Mitigation • Supplier Development • DFM/DFA • Problem Solving • Design Review • IATF 16949, ISO 9001

Software: Minitab, Plex, MES, Teamcenter/Enovia, MS Office, Gage Pack, SQL, Tableau, Power BI

TECHNICAL & MANUFACTURING EXPERTISE

Injection Molding • Stamping (Transfer & Progressive) Roll Forming • Extrusion • Welding (MIG, TIG, RSW) • Machining • General Assembly • Powder Coating • E-Coating • Heat Treat • EV Battery Components • Interior/Exterior Trim • Poka-Yoke Development

Certifications & Training: ASQ CPQA ,8D Problem Solving, Six Sigma, MSA, SQL